Business Overview & Goal

Sales & Revenue Performance Analysis

(Northwind Database)

Business Context

- The company sells products across multiple categories to different customers.
- Transactions occur over time with varying price and discounts.

Goal of the Analysis

- Move from sales counts to revenue driven insights
- Understand what generates money, who generates it, and when it happens
- Support better business decision using data

Business Questions & Analytical Focus

Key Business Question

- Which products generate the height revenue?
- Which product categories contribute most to total revenue?
- How does revenue change over time (monthly trends)
- How can customers be segmented based on revenue contribution?

Data Understanding & Tables Used

Dataset Overview

Northwind Sample Dataset

Key Tables used

- Customers – Customer details and identifier
- Orders – Order – level information and order dates
- OrderDetails – Product – level transaction details
- Products – Product information and pricing
- Categories – Product category classification

Data Relationship

Customer → Orders → OrderDetails → Products → Categories

Revenue Calculation Logic & Metrics

Revenue Definition

Revenue represents the actual money earned from each transaction after discounts.

Revenue Formula: UnitPrice * Quantity

Key Metrics Created

- Total Revenue
- Revenue by Product
- Revenue by Category
- Revenue by Customer
- Monthly Revenue

SQL Concepts & Techniques Used

SQL Concepts Applied

- Multi-table JOINs to combine transactional and master data
- Aggregation functions: SUM( )
- GROUP BY for summarizing revenue
- ORDER BY for ranking results
- Date functions: YEAR( ), MONTH( )
- Conditional logic using CASE WHEN
- Common Table Expressions (CTEs) for modular queries

Top Revenue-Generating Products

Business Question 1

Which products generate the highest revenue?

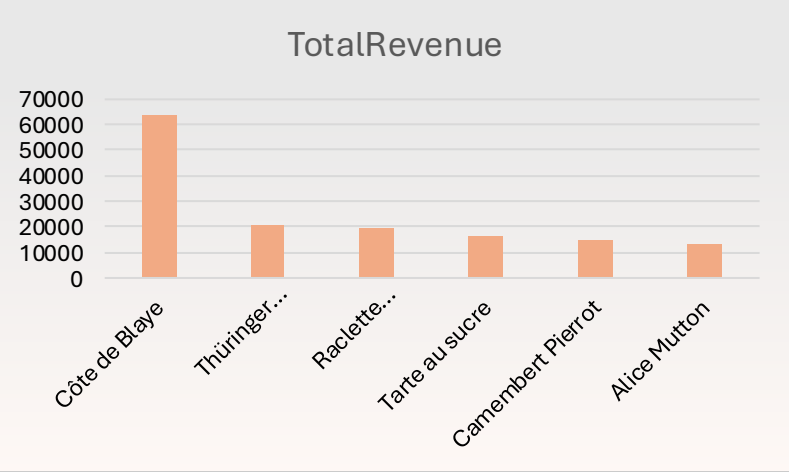
Key Findings

- Cote de Blaye is the top revenue generating product, with a total revenue of 63,096
- The second and third products (Thuringer Rostbratwurst and Raclette Courdavault) generate significantly less revenue
- Revenue contribution is highly concentrated among a small number of products
- High products price alone does not guarantee high revenue – sales volume matters

Insight Summary

- A small subset of products drives a large portion of total revenue
- Focusing on top-performing products can maximize business impact

```
SELECT
   p.ProductName,
   SUM(p.Price * od.Quantity) AS TotalRevenue
FROM Northwind.dbo.Products p
JOIN Northwind.dbo.OrderDetails od
   ON p.ProductID = od.ProductID
GROUP BY p.ProductName
ORDER BY TotalRevenue DESC;
```



Business Impact and Recommendations

Business Impact

- Revenue is not evenly distributed across products
- A small number of products generate most of the revenue
- Product price alone does not determine success - sales volume is critical

Recommendations

- **Prioritize top revenue - generating products**
  - Maintain sufficient inventory
  - Avoid stock-outs of high-impact products
- **Focus marketing efforts on high-performing products**
  - Promote best-selling items
  - Bundle or cross-sell related products
- **Review low-revenue products**
  - Evaluate pricing and promotion strategies
  - Consider optimization or rationalization

Revenue Contribution by Product Category

Business Question

Which product categories contribute the most to total revenue?

Analytical Path

Categories → Products → OrderDetails

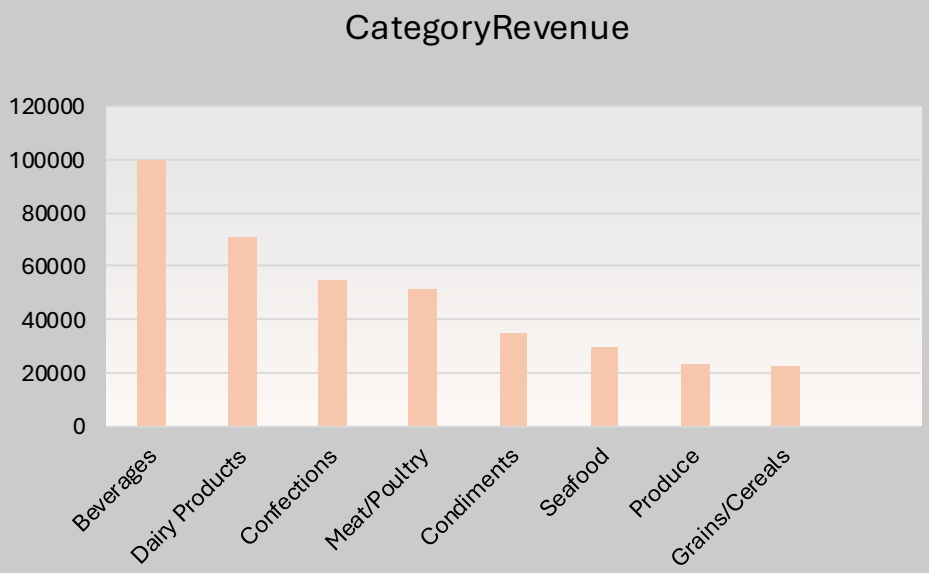
Key Findings

- Beverages is the highest revenue-generating category (99,699)
- Dairy Products and Confections are the next top contributors
- Revenue contribution varies significantly across categories

Insight Summary

- Revenue is concentrated in a small number of categories
- Category-level analysis provides clearer strategic direction than individual products

```sql
SELECT
    c.CategoryName,
    SUM(p.Price * od.Quantity) AS CategoryRevenue
FROM Northwind.dbo.Categories c
JOIN Northwind.dbo.Products p
    ON c.CategoryID = p.CategoryID
JOIN Northwind.dbo.OrderDetails od
    ON p.ProductID = od.ProductID
GROUP BY c.CategoryName
ORDER BY CategoryRevenue DESC;
```



Monthly Revenue Trend

Business Question

How does revenue change over time on a monthly basis?

Analytical Path

Products → OrderDetails → Orders

Key Findings

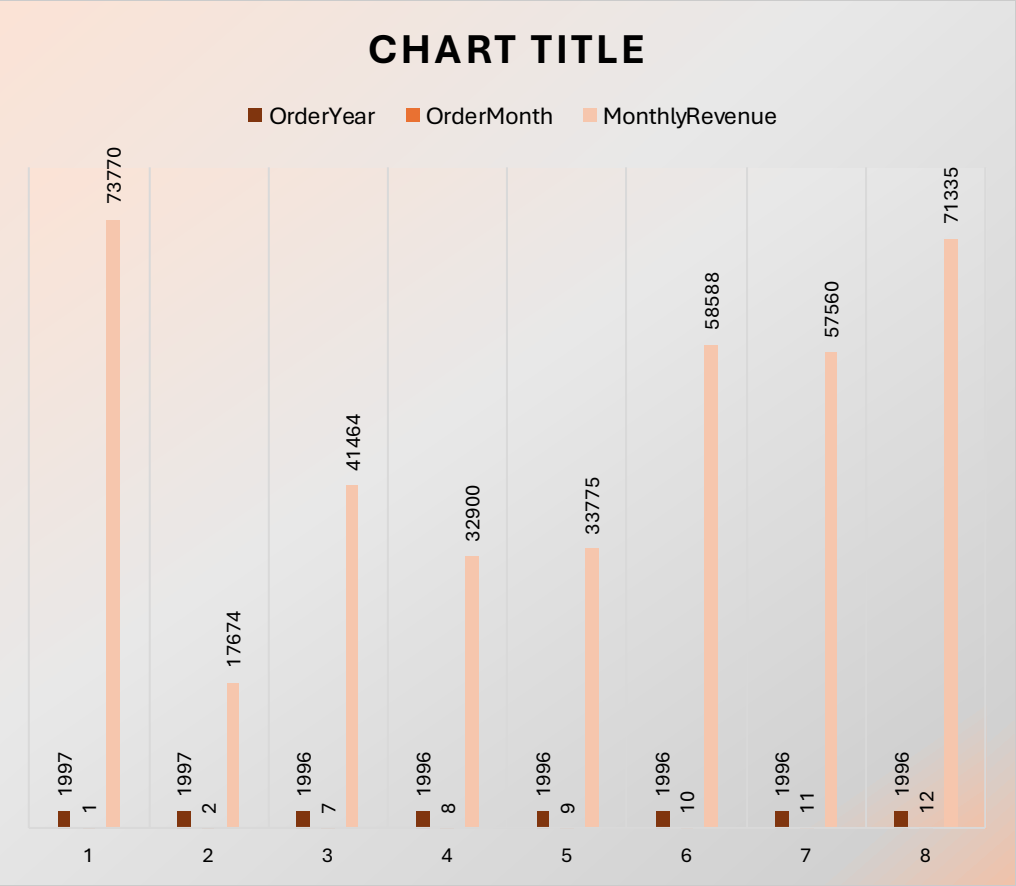
- Revenue shows clear month-to-month variation
- Revenue increased significantly during Q4 1996 (October–December)
- December 1996 and January 1997 are the highest revenue months
- Revenue dropped sharply in February 1997

Insight Summary

- Revenue follows a seasonal pattern
- Certain months consistently generate higher revenue
- Time-based analysis supports better forecasting and planning

SQL QUERY

```
SELECT
  YEAR(o.OrderDate) AS OrderYear,
  MONTH(o.OrderDate) AS OrderMonth,
  SUM(p.Price * od.Quantity) AS MonthlyRevenue
FROM Northwind.dbo.Products p
JOIN Northwind.dbo.OrderDetails od
  ON p.ProductID = od.ProductID
JOIN Northwind.dbo.Orders o
  ON od.OrderID = o.OrderID
GROUP BY YEAR(o.OrderDate),
MONTH(o.OrderDate)
ORDER BY OrderYear, OrderMonth;
```



Customer Revenue Segmentation

Business Question

How can customers be segmented based on their revenue contribution?

Analytical Path

Customers → Orders → OrderDetails → Products

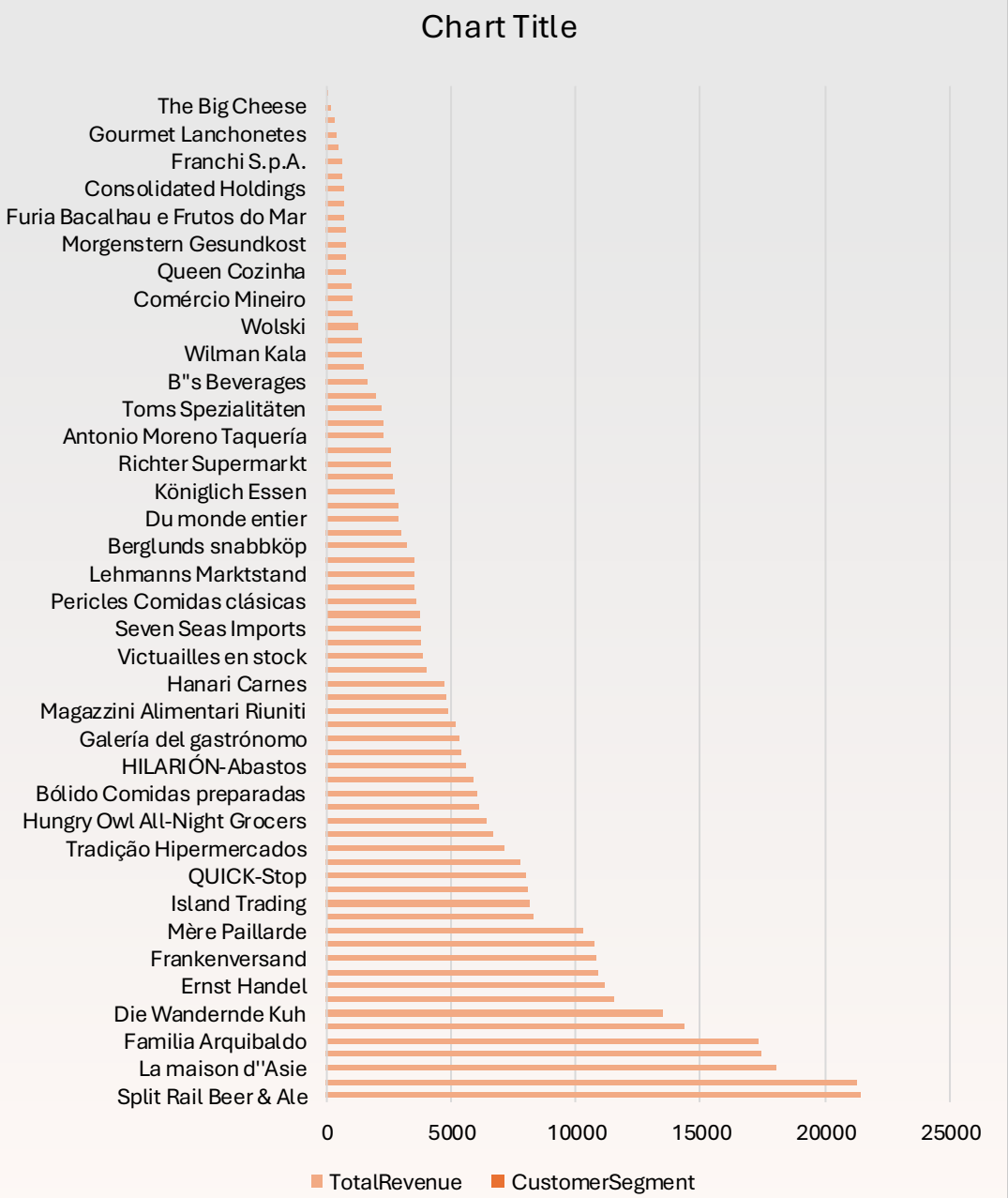
Segmentation Logic

- Customers are grouped based on total revenue contribution:
- **High Value**: Revenue ≥ 21,000
- **Medium Value**: Revenue between 10,000 and 20,999
- **Low Value**: Revenue < 10,000

Key Findings

- Only a small number of customers fall into the High Value segment
- The Medium Value segment contains a moderate number of customers
- Low Value customers form the largest group by count but contribute the least to total revenue

```
WITH CustomerRevenue AS (
SELECT
c.CustomerName,
SUM(od.Quantity * p.Price) AS TotalRevenue
FROM Northwind.dbo.Customers c
JOIN Northwind.dbo.Orders o ON c.CustomerID = o.CustomerID
JOIN Northwind.dbo.OrderDetails od ON o.OrderID =
od.OrderID
JOIN Northwind.dbo.Products p ON od.ProductID = p.ProductID
GROUP BY c.CustomerName
)
SELECT
CustomerName,
TotalRevenue,
CASE
WHEN TotalRevenue >= 21000 THEN 'High Value'
WHEN TotalRevenue >= 10000 THEN 'Medium Value'
ELSE 'Low Value'
END AS CustomerSegment
FROM CustomerRevenue
ORDER BY TotalRevenue DESC;
```



Key Takeaways & Business Recommendations

Key Takeaways

- Revenue is highly concentrated among a small set of products and customers
- Category-level analysis reveals clear revenue leaders
- Monthly trends indicate seasonality in sales performance
- Customer revenue follows a long-tail distribution

Business Recommendations

- Prioritize high-revenue products and categories for inventory and promotions
- Focus on retention strategies for High Value customers
- Upsell and cross-sell to Medium Value customers to increase lifetime value
- Use seasonal trends to optimize marketing and supply planning